

5-amaliy mashg'ulot. Xususiyatlar va moslashtirish

2026–2027 o'quv yili

Muhitni sozlash. Python 3.11 yoki 3.12, faqat CPU. Versiyalar qat'iy belgilangan (requirements.txt): numpy==2.4.4, scipy==1.17.1, pandas==3.0.2, scikit-learn==1.8.0, scikit-image==0.26.0, matplotlib==3.10.9, pillow==12.2.0, opencv-python-headless==4.13.0.92, torch==2.14.0, torchvision==0.29.0. OpenCV 4.13 da SIFT (cv2.SIFT_create) va ORB (cv2.ORB_create) asosiy paketda, patentsiz.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
python -m pip install --upgrade pip && pip install -r requirements.txt
```

Mavzu. Tuzilma matritsasi, Harris va Shi–Tomasi detektorlari, masshtab fazosi (LoG, DoG), SIFT, FAST, BRIEF, ORB, Hamming masofasi, Lowe nisbat testi, RANSAC va homography (DLT), HOG.

Maqsadlar. Talaba tuzilma matritsasi, Harris javobi R va xos qiymatlarni qo'lda hisoblaydi hamda tekis soha/chegara/burchakni ajratadi; SIFT, FAST, ORB bosqichlarini kichik misolda bajaradi; RANSAC iteratsiyalar sonini va homography ni hisoblaydi; OpenCV'da detektor + deskriptor + moslashtirish + RANSAC zanjirini quradi va uni ma'lum geometrik almashtirish bo'yicha sonli tekshiradi.

Baholash (100 ball). 1–16-topshiriqlar: har biri 2 ball (32). 17–30-topshiriqlar (kod): 26, 27, 29 (★, amaliy holatlar) — har biri 8 ball (24), qolgan 11 tasi — 4 ball (44). Kod xatosiz ishlashi, seed=42 bilan takrorlanishi va ko'rsatilgan formatda chiqarishi shart.

Kelishuv. u — ustun (o'ngga), v — qator (pastga). Oyna w ichida $M = \sum w \begin{pmatrix} I_u^2 & I_u I_v \\ I_u I_v & I_v^2 \end{pmatrix} = \begin{pmatrix} A & C \\ C & B \end{pmatrix}$, $R = \det M - k(\text{tr } M)^2$ ($k = 0,04$, agar boshqacha aytilmasa), $R_{ST} = \lambda_{\min}$, $\lambda_{1,2} = \frac{A+B}{2} \pm \sqrt{\left(\frac{A-B}{2}\right)^2 + C^2}$.

A qism. Qo'lda hisoblash, keltirib chiqarish, nazariya

- Tuzilma matritsasi.** 3×3 oyna, $w \equiv 1$. a) $I_u = \begin{pmatrix} 4 & 6 & 0 \\ 5 & 8 & 2 \\ 0 & 2 & 1 \end{pmatrix}$, $I_v = \begin{pmatrix} 0 & 2 & 5 \\ 1 & 3 & 6 \\ 4 & 6 & 2 \end{pmatrix}$. b) Har bir pikselda $\nabla I = c_i(3, -4)$, $c = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$. Ikkalasi uchun $A, B, C, \lambda_{1,2}, R, R_{ST}$ va holat (b da e_1 ham).
- Harris javobi va nisbiy ostona.** $M_1 = \begin{pmatrix} 8 & 2 \\ 5 & 5 \end{pmatrix}$, $M_2 = \begin{pmatrix} 20 & 10 \\ 10 & 5 \end{pmatrix}$, $M_3 = \begin{pmatrix} 0,5 & 0,1 \\ 0,1 & 0,4 \end{pmatrix}$: det, tr, $R, \lambda_{1,2}$. Ostona $R > 0,01 \max R$ bo'lsa qaysilari burchak deb olinadi?
- Piksellardan Harris gacha.** 7×7 tasvirda $I(u, v) = 20$, agar $u \geq 3$ va $v \leq 3$, aks holda 0. $I_u = \frac{1}{2}[I(u+1, v) - I(u-1, v)]$, I_v shunga o'xshash; $(3, 3)$ markazli 3×3 oyna, $w \equiv 1$. Oynadagi I_u, I_v , so'ng $A, B, C, \lambda_{1,2}, R$ va e_1 .
- Ishora chegarasi ρ_k .** $R = \lambda_1^2[\rho - k(1 + \rho)^2]$ dan $R > 0 \iff \rho > \rho_k$, $\rho_k = \frac{1-2k-\sqrt{1-4k}}{2k}$ ni keltirib chiqaring. $k = 0,04$ va $k = 0,05$ uchun ρ_k ; $\lambda = (100, 5)$ uchun ikkala k da R . Xulosa?
- Invariantlik.** a) $I \rightarrow aI + b$ da M va R qanday o'zgaradi ($a = 2, b = 30$)? b) 90° burish $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ uchun $M' = QM_1Q^T$ ni hisoblang va det, tr o'zgarimasligini tekshiring. c) Nega Harris masshtabga invariant emas? Apertura muammosini bir jumlada izohlang.
- Masshtab.** Radiusi r bo'lgan doira markazida $\sigma^2|\Delta L| = \frac{r^2}{\sigma^2}e^{-r^2/(2\sigma^2)}$ ekanidan $\sigma^* = r/\sqrt{2}$ ni chiqaring; $r = 9$ va tasvir 1,5 marta kattalashtirilgandan keyin σ^* . SIFT oktavasi: $\sigma_0 = 1,6, s = 3, k = 2^{1/s}$ — barcha Gauss masshtablari (3 xona), DoG tasvirlari soni, keyingi oktava qaysi σ dan boshlanadi?
- SIFT filtrlari.** Chegara sharti $\frac{(\text{tr } H)^2}{\det H} < \frac{(r+1)^2}{r}$, $r = 10$. $H_1 = \begin{pmatrix} -12 & 2 \\ 2 & -1,5 \end{pmatrix}$, $H_2 = \begin{pmatrix} -6 & 1 \\ 1 & -4 \end{pmatrix}$, $H_3 = \begin{pmatrix} 5 & 3 \\ 3 & -2 \end{pmatrix}$: qaysi nomzod qoladi va nega? $|D(\hat{z})| = 0,02$ bo'lsa-chi?
- Yo'nalish cho'qqisi.** 36 chelakli (10°) gistogrammada $h(110^\circ) = 9, h(120^\circ) = 14, h(130^\circ) = 12$. Parabola bilan aniqlangan yo'nalish. Boshqa lokal cho'qqilar: 250° da 12, 300° da 11 — qaysi biri qo'shimcha kalit nuqta beradi (80% qoidasi)?
- Deskriptorni normallashtirish.** Soddalashtirilgan deskriptor $d = (6, 3, 2, 1, 1, 1)$: L_2 normallashtirish, 0,2 da qirg'ish, qayta normallashtirish (4 xona). Qirg'ishning ma'nosi nima? To'liq SIFT deskriptori uzunligi qanday chiqadi?
- FAST.** $I_p = 80, t = 15$, aylana (1–16): 100, 110, 98, 120, 101, 85, 60, 50, 55, 62, 70, 90, 99, 130, 115, 97. FAST-9 va FAST-12 bo'yicha burchakmi (aylana yopiq, 16 dan 1 ga o'tish mumkin)? 1, 5, 9, 13-piksellar tezkor testi nima beradi?
- BRIEF va Hamming.** 16 bitli deskriptorlar 0xB3C5 va 0x9A4D: XOR va Hamming masofasi. 256 bitli ikki tasodifiy (bog'liqsiz) ORB deskriptori uchun masofaning kutilmasi. Nega ORB uchun L_2 emas?

- 12. ORB yo'nalishi.** $u, v \in \{-1, 0, 1\}$ (v pastga), $m_{pq} = \sum u^p v^q I$, $\theta = \text{atan2}(m_{01}, m_{10})$. a) $I = \begin{pmatrix} 0 & 2 & 4 \\ 0 & 3 & 6 \\ 1 & 2 & 8 \end{pmatrix}$; b) faqat chap-yuqori piksel 9, qolganlari 0.
- 13. Nisbat testi.** (d_1, d_2) : (0,20;0,50), (0,35;0,40), (0,30;0,42), (0,10;0,12), (0,60;0,95), $\tau = 0,75$. Qaysilari qabul qilinadi? Nega eng kichik $d_1 = 0,10$ bo'lgan juft rad etiladi? Cross-check nima?
- 14. RANSAC iteratsiyalari.** $N \geq \frac{\ln(1-p)}{\ln(1-\omega^s)}$ ni keltirib chiqaring. $p = 0,99$, $\omega = 0,4$ da $s = 2$ (to'g'ri chiziq), $s = 3$ (affin), $s = 4$ (homography), $s = 8$ (fundamental matritsa) uchun N .
- 15. Homography.** $H = \begin{pmatrix} 2 & 0 & 10 \\ 0 & 2 & -4 \\ 0,01 & 0 & 1 \end{pmatrix}$. (50, 20) nuqtaning tasviri; (50, 20) $\rightarrow$ (74, 25) moslik $\varepsilon = 3$ da inliermi? (1, 2) $\rightarrow$ (3, 5) moslik uchun DLT ning ikki qatorini yozing. Homography, affin va o'xshashlik almashtirishlari uchun erkinlik darajalari va minimal moslik soni.
- 16. HOG.** $D = \left(\frac{W}{c} - b + 1\right) \left(\frac{H}{c} - b + 1\right) b^2 n_b$. 96×160 ($W \times H$) oyna, $b = 2$, $n_b = 9$: $c = 8$ va $c = 16$. Bilinear ovoz (markazlar $10^\circ, 30^\circ, \dots, 170^\circ$): $\theta = 137^\circ$, $M = 8$ va $\theta = 5^\circ$, $M = 4$ — qaysi chelaklarga qancha?

B qism. Dasturlash (Python, OpenCV, scikit-image)

Umumiy kod (skimage.data tasvirlari paket ichida; tashqi fayllar faqat 27-topshiriqda):

```
import math, urllib.request
from pathlib import Path
import numpy as np, cv2
from scipy import ndimage as ndi
from skimage import data
from skimage.feature import hog
SEED = 42
URL = "https://raw.githubusercontent.com/opencv/opencv/4.x/samples/data/"

def fetch(name): # ./data/ ga yuklab, keshlaydi
    p = Path("data") / name; p.parent.mkdir(exist_ok=True)
    if not p.exists(): urllib.request.urlretrieve(URL + name, p)
    return p

def gray(name): # skimage.data -> kulrang uint8
    a = getattr(data, name)()
    return cv2.cvtColor(a, cv2.COLOR_RGB2GRAY) if a.ndim == 3 else a

to_xy = lambda kps: np.float64([k.pt for k in kps]).reshape(-1, 2)
apply_affine = lambda M, p: p @ M[:, :2].T + M[:, 2] # M: 2x3
def apply_h(H, p): # H: 3x3
    q = np.c_[p, np.ones(len(p))] @ H.T
    return q[:, :2] / q[:, 2:]
```

Har bir topshiriq alohida funksiya bo'lib, natijani ko'rsatilgan formatda chiqaradi; rasmlar ./out/ ga. Hujjatlar: https://docs.opencv.org/4.x/dc/d0d/tutorial_py_features_harris.html, https://docs.opencv.org/4.x/d4/d8c/tutorial_py_shi_tomasi.html, https://docs.opencv.org/4.x/da/df5/tutorial_py_sift_intro.html, https://docs.opencv.org/4.x/df/d0c/tutorial_py_fast.html, https://docs.opencv.org/4.x/d1/d89/tutorial_py_orb.html, https://docs.opencv.org/4.x/dc/dc3/tutorial_py_matcher.html, https://docs.opencv.org/4.x/d1/de0/tutorial_py_feature_homography.html, <https://scikit-image.org/docs/stable/api/skimage.feature.html#skimage.feature.hog>, https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.gaussian_laplace.html.

- 17. Harris qo'lda.** `harris_manual(img, k=0.04, block=3) -> np.ndarray`: `cv2.Sobel (CV_32F)`, `cv2.boxFilter(..., normalize=False)`, $R = AB - C^2 - k(A + B)^2$. Tasvir: `data.checkerboard()/255 (float32)`. `cv2.cornerHarris(img, 3, 3, 0.04)` bilan korrelyatsiya (6 xona) va R/R_{cv} nisbatining medianasi. NMS: $R == \text{ndi.maximum_filter}(R, \text{size}=7)$ va $R > 0,1 \max R$; lokal maksimum piksellari soni va ularning 8-bog'langan platolari soni (burchaklar), birinchi 3 plato markazi. Nisbat 12^4 ga teng ekanini tushuntiring.
- 18. Xos qiymatlar xaritasi.** `cv2.cornerEigenValsAndVecs(camera/255, blockSize=5, ksize=3)` dan λ_1, λ_2 ; $\tau = 0,01 \max \lambda_1$: tekis ($\lambda_1 < \tau$), burchak ($\lambda_2 \geq \tau$), qolgani chegara — ulushlar (4 xona). λ lardan hisoblangan R ni `cv2.cornerHarris(f, 5, 3, 0.04)` bilan, $\min(\lambda_1, \lambda_2)$ ni `cv2.cornerMinEigenVal(f, 5, 3)` bilan solishtiring (maksimal farq).
- 19. Shi-Tomasi va Harris.** `cv2.goodFeaturesToTrack(camera, maxCorners=200, qualityLevel=0.01, minDistance=10, blockSize=3)` va shu parametrlar + `useHarrisDetector=True, k=0.04`. Nuqtalar soni, Shi-Tomasi nuqtalaridan Harris nuqtasiga ≤ 3 px bo'lganlar soni va ulushi; rasm `out/t19_shi_tomasi.png`.
- 20. Takrorlanuvchanlik.** `repeatability(pts_a, pts_b, M, shape, tol=3.0, margin=10)`: A nuqtalarini M (2×3) bilan ko'chirib, kadr ichida (chetdan 10 px) qolganlaridan B da ≤ 3 px topilganlari ulushi. `camera, cv2.getRotationMatrix2D((256, 256), ang, 1.0)`, `warpAffine, ang = 15, 30, 45`. Detektorlar: Shi-Tomasi va Harris (`maxCorners=300, qualityLevel=0.01, minDistance=7`), `cv2.SIFT_create(nfeatures=300)`.

- 21. Xarakteristik masshtab.** $(8r + 1)^2$ tasvirda radiusi $r = 4, 8, 16$ bo'lgan disk (1/0); markazda $\sigma^2 | \text{ndi.gaussian_laplace}(\text{disk}, \sigma, \text{mode}='constant') |$ ni $\sigma = 1,00:0,05:20,00$ bo'yicha maksimalashtiring. σ^* va $r/\sqrt{2}$ (2 xona).
- 22. DoG va LoG.** `camera` (`float64`), $\sigma = 2$, $k \in \{1, 6; 2^{1/3}; 1, 1\}$: $D = G_{k\sigma} * I - G_{\sigma} * I$ (`ndi.gaussian_filter`) va $(k - 1)\sigma^2 \text{ndi.gaussian_laplace}(I, \sigma)$ korrelyatsiyasi (4 xona) hamda eng kichik kvadratlar masshtabi $\beta = \langle D, N \rangle / \langle N, N \rangle$ (3 xona). $k \rightarrow 1$ da nima bo'ladi?
- 23. SIFT statistikasi.** `cv2.SIFT_create()` bilan `gray('astronaut')`: nuqtalar soni, deskriptor shakli va turi, L_2 normasi o'rtachasi, maksimal komponent; oktavalar bo'yicha taqsimot (oktava = `k.octave` & 255, ≥ 128 bo'lsa -256), `k.size` medianasi va maksimumi. Nega norma 512?
- 24. FAST.** `cv2.FastFeatureDetector_create(threshold=t, nonmaxSuppression=...)`, `camera`, $t = 10, 20, 40$: NMS bilan va NMS siz nuqtalar soni va nisbati (2 xona).
- 25. ORB va Hamming.** `hamming(a, b) -> int (np.unpackbits(np.bitwise_xor(a, b)).sum())`. `camera` va uning 20° ga burilgani (`getRotationMatrix2D((256, 256), 20, 1.0)`), `ORB_create(nfeatures=500)`, `BFMatcher(NORM_HAMMING, crossCheck=True)`; mosliklarni (`distance, queryIdx`) bo'yicha tartiblang. Deskriptor shakli, mosliklar soni, qo'lda hisoblangan masofa BF bilan barcha mosliklarda tengmi, min va median masofa, to'g'ri (≤ 3 px) mosliklar soni va eng yaxshi 50 tadan to'g'rilari.
- 26. * ORB + RANSAC bilan tasvirni tekislash.** $g = \text{camera}$, $H_{\text{true}} = \begin{pmatrix} 0,9 & 0,15 & 30 \\ -0,1 & 0,95 & 40 \\ 2 \cdot 10^{-4} & 10^{-4} & 1 \end{pmatrix}$, `g2` = `warpPerspective(g, H_true, (512, 512)) + N(0, 3^2)` shovqin (`default_rng(42)`), `uint8` ga qirqish. `ORB_create(nfeatures=2000)`, `knnMatch(k=2)`; $\tau = 0,75$ va $\tau = 1,0$ (nisbat testisiz, eng yaqin qo'shni). Uch baho: barcha mosliklarda LS (`findHomography(src, dst, 0)`); `cv2.setRNGSeed(42)` va `findHomography(src, dst, cv2.RANSAC, 3.0, maxIters=2000, confidence=0.999)`; RANSAC inlierlarida qayta LS. Xato: 10×10 nazorat to'rida ($[32, 480]^2$) $\|H(x) - H_{\text{true}}(x)\|$ ning o'rtachasi/maksimumi (2 xona). $\tau = 1,0$, RANSAC+LS bilan `g2` ni orqaga tekislab, umumiy sohada o'rtacha [farq]; `out/t26_aligned.png`.
- 27. * Obyektni sahnada topish (SIFT).** `box.png` va `box_in_scene.png`: <https://raw.githubusercontent.com/opencv/opencv/4.x/samples/data/box.png>, https://raw.githubusercontent.com/opencv/opencv/4.x/samples/data/box_in_scene.png (kulrang). SIFT, `BFMatcher(NORM_L2).knnMatch(k=2)`, $\tau = 0,75$, `setRNGSeed(42)`, RANSAC (5,0 px). Nuqtalar, nisbat testidan o'tganlar, inlierlar; quti burchaklarining sahnadagi koordinatalari (butun) va to'rtburchak yuzasi (`cv2.contourArea`); `out/t27_box_found.png`.
- 28. Nisbat testi ostonasi.** `gray('astronaut')` va `warpAffine` bilan `getRotationMatrix2D((256, 256), 25, 0.8)`; SIFT, `knnMatch(k=2)`; $\tau = 0,6; 0,7; 0,8; 0,9; 1,0$ (1,0 — testsiz): mosliklar, to'g'ri (≤ 3 px), aniqlik (3 xona).
- 29. * RANSAC noldan.** `dlt_homography(src, dst) -> H` (Hartley normallashtirishi, SVD) va `ransac_homography(src, dst, thr=3.0, p=0.99, seed=42, max_iter=5000) -> (H, inliers, iters)`: 4 nuqta, eng ko'p inlierli model, adaptiv $N = \lceil \ln(1 - p) / \ln(1 - \omega^4) \rceil$, oxirida inlierlarda DLT, qayta baholash va yana DLT. Ma'lumot: `rng = default_rng(42)`; $H_{\text{true}} = \begin{pmatrix} 1,1 & 0,2 & 15 \\ -0,15 & 0,95 & 30 \\ 3 \cdot 10^{-4} & -2 \cdot 10^{-4} & 1 \end{pmatrix}$, `src = rng.uniform(0, 500, (200, 2))`, `dst = apply_h(H_true, src) + rng.normal(0, 1, (200, 2))`, `idx = rng.choice(200, 120, replace=False)`, `dst[idx] = rng.uniform(0, 600, (120, 2))` (shu tartibda). Iteratsiyalar va nazariy N ($\omega = 0,4$), inlierlar, haqiqiy inlierlardan topilgani, soxtalar; $[0, 500]^2$ burchaklarida maksimal xato: RANSAC va barcha nuqtalardagi DLT (3 va 1 xona).
- 30. HOG.** Oyna: `cv2.resize(gray('astronaut')[30:330, 130:280], (64, 128))`. `hog(..., orientations=9, pixels_per_cell=(c, c), cells_per_block=(2, 2), block_norm='L2-Hys')`, $c = 8, 16$ — uzunliklar; `cv2.HOGDescriptor((64, 128), (16, 16), (8, 8), (8, 8), 9)` uzunligi. 64×64 naqshlar $\sin(2\pi v/8)$, $\sin(2\pi u/8)$, $\sin(2\pi(u+v)/8)$ uchun bitta katak (`pixels_per_cell=(64, 64)`, `cells_per_block=(1, 1)`, `block_norm='L2'`) gistogrammasining hukmron chelagi. Natijani gradient va chegara yo'nalishi orqali izohlang.